

Notes: Welch (JF, 1992)

Sequential Sales, Learning, and Cascades

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1 Big picture

- **Question.** What happens when IPO shares or other new securities are sold sequentially and later investors can learn from earlier investors' purchase decisions?
- **Main idea.** Sequential selling creates an informational externality: investors approached later observe whether earlier investors bought or abstained. Their actions can therefore reveal information, even when their private signals are not directly observable.
- **Core mechanism.** Once enough early investors buy, later investors may rationally ignore their own private information and buy because they infer favorable earlier information. Conversely, once enough early investors abstain, later investors may ignore their private information and abstain. These are informational cascades.
- **IPO application.** Underwriters usually approach investors over time. Early demand, weak demand, or the absence of announced interest can therefore affect later investors' beliefs and change the success or failure of the offering.
- **Main theoretical result.** Under cascades, demand can become extremely elastic. A risk-neutral uninformed issuer may optimally underprice enough to avoid offering failure, not because of a standard winner's curse among marginal investors, but because early investor participation must be induced.
- **Winner's curse result.** Rationing and winner's curse logic can exist, but it is not the central mechanism. In this model, the issuer prices to start a favorable cascade, not to compensate marginal investors for being allocated bad issues.
- **Path dependence.** In simple path dependence, later investors observe earlier investors' signals; in cascades, they observe only earlier actions. The difference matters because action observation can cause investors to stop using their own information.
- **Inside information result.** If issuers have private information, high-quality issuers can signal by choosing a high enough price that they risk failure. Offering failure can therefore be a costly but informative signal by issuers with good inside information.
- **Underwriter implication.** Underwriters may have incentives to reduce communication among investors by segmenting the sales effort. This is counterintuitive because more communication sounds like better information aggregation, but too much communication can create harmful path dependence from the issuer's perspective.
- **Empirical predictions.** Offerings should tend to succeed or fail quickly; underpricing should rise with uncertainty; local offerings are more vulnerable to cascades and failures; and national offerings may be less likely to fail when issuer inside information matters.

2 Incremental contribution of the paper

2.1 Contribution relative to Rock-style winner's curse models

- Rock-type IPO models explain underpricing through rationing and the winner's curse: uninformed investors receive more shares in bad offerings and must be compensated.
- Welch's contribution is to show that underpricing can arise even when the winner's curse is not the relevant force.
- The new force is **sequential information transmission**: later investors condition on earlier purchase decisions.
- Underpricing is used to induce early investors to buy and thereby start a favorable cascade.

Interpretation: The paper shifts the focus from allocation after demand is known to belief formation while demand is being created. This is a different margin of IPO design.

2.2 Contribution relative to static demand-curve explanations

- Static downward-sloping demand stories treat demand for the issue as a contemporaneous function of price.
- Welch shows that demand itself can be path dependent: the same price and the same information distribution can produce different outcomes depending on the order of investors' signals and actions.
- This makes IPO demand much more fragile than in a static model.

Interpretation: The paper's incremental insight is that demand is not just downward sloping; it can be dynamically self-reinforcing because investors learn from one another's actions.

2.3 Contribution relative to information cascade theory

- Information cascades are usually presented in abstract social-learning environments.
- Welch applies the logic to a concrete financial-market institution: sequential IPO sales.
- The model links cascades to underpricing, offering failure, underwriter sales strategy, pre-selling, and market segmentation.

Interpretation: The paper helps bring social learning into corporate finance by showing how sequential sales can convert small early signals into large offering outcomes.

2.4 Contribution relative to signaling theories of IPOs

- Signaling models often use underpricing or subsequent equity offerings as the signaling device.
- Welch introduces a setting in which **offering failure risk** can itself be part of signaling.
- A high-quality issuer may choose a high price, risk failure, and separate from low-quality issuers because the high-quality issuer is more likely to face favorable outside information.

Interpretation: The paper gives a new interpretation of withdrawn or failed offerings: failure can occur not because the issuer is bad, but because a high-quality issuer rationally gambled on a high price and unfavorable early investor signals occurred.

2.5 Contribution to the theory of underwriter organization

- The paper provides a rationale for underwriters' control over the sales process and information flow among investors.
- If investors communicate perfectly, the issuer faces the full informational disadvantage of the market.
- If investors observe only actions, cascades can sometimes help the issuer by creating self-reinforcing demand.
- Segmenting sales across investors or markets can therefore be valuable.

Interpretation: Underwriter networks and selling procedures are not just marketing technology. They are part of the information structure of the offering.

3 Model primitives and setup

3.1 Issuer, investors, and the offering price

- There is one issuer selling a new security, interpreted as an IPO share.
- There are n potential investors.
- The issuer approaches investors sequentially.
- Each investor can buy at most one share or abstain.
- The issuer must choose one fixed offer price P for all investors.
- Investors buy if the expected aftermarket value conditional on their information is at least the offer price.

Interpretation: The fixed-price constraint is crucial. The issuer cannot continuously adjust price as early demand is revealed, so early investor behavior affects later beliefs rather than just the price.

3.2 Firm value and type transformation

The aftermarket value of a share is uncertain. Let V be the efficient aftermarket value. The paper transforms the value interval into a normalized type:

$$\theta \in [0, 1], \quad V = (1 - \theta)V_L + \theta V_H. \quad (1)$$

A type $\theta = 0$ firm is worth V_L , a type $\theta = 1$ firm is worth V_H , and a type $\theta = 1/2$ firm is worth the midpoint.

Interpretation: Normalizing the value to $[0, 1]$ allows the offering price and posterior expected value to be stated as probabilities or fractions.

3.3 Investor signals

Each investor observes one private signal:

$$s_i \in \{H, L\}. \quad (2)$$

The probability of an H signal equals the underlying type:

$$\Pr(s_i = H \mid \theta) = \theta. \quad (3)$$

Interpretation: A high-value issue is more likely to generate favorable investor signals. Aggregating many signals can therefore reveal the firm's value.

3.4 Posterior valuation

With a uniform prior on θ , if k of n observed signals are H , the posterior mean is

$$E(\theta \mid k, n) = \frac{k + 1}{n + 2}. \quad (4)$$

Therefore the highest normalized price at which investors buy after observing k high signals in n total signals is

$$p^{\max}(k, n) = \frac{k + 1}{n + 2}. \quad (5)$$

Interpretation: These simple Bayesian updating formulas generate the thresholds in the game trees. One H signal gives posterior value $2/3$; one L signal gives posterior value $1/3$.

3.5 Communication regimes

The paper compares three information regimes:

1. **Perfect communication:** all investors' signals are observed or communicated.
2. **Simple path dependence:** each later investor observes the earlier investors' signals, but only in the order the underwriter approaches them.
3. **Cascades:** each later investor observes only earlier investors' actions, not their private signals.

Interpretation: The difference between observing signals and observing actions is the source of cascades. Actions are coarser than signals, so later investors may stop learning the true signal history.

4 Core mechanisms and results

4.1 Perfect communication benchmark

With perfect communication, the issuer can choose a price that succeeds if the number of high signals exceeds a threshold. If the issuer wants the issue to sell whenever at least k investors receive high signals, the relevant price is approximately $p^{\max}(k, n)$.

The issuer chooses a trade-off between:

- a high price that raises proceeds if the issue succeeds;
- a lower probability of success because fewer signal realizations induce investors to buy.

Interpretation: This benchmark resembles a static demand problem. Investors collectively know the issue's value, and the issuer optimizes against that informationally superior market.

4.2 Theorem 1: underpricing with perfect communication

Theorem 1 states that a risk-neutral uninformed issuer never prices above the value of the average type, that successful offerings are underpriced on average, and that expected issuer utility is bounded in large markets.

Read:

- If the offering succeeds, investors must have sufficiently favorable aggregate information.
- Conditional on success, the aftermarket value is therefore above the offer price on average.
- Successful offerings exhibit positive initial returns.
- The average underpricing of successful offerings increases with ex-ante uncertainty.

Economic message: Underpricing can arise from the issuer's informational disadvantage even before cascades are introduced. The later contribution of the cascade model is to show an additional and different reason for underpricing.

4.3 Simple path dependence

In simple path dependence, investors approached later observe earlier investors' private signals. Because the order of investors matters, the issuer's proceeds can be path dependent.

For example, with two investors and a price of $2/5$, observing H then L can produce a different outcome from observing L then H , because the first investor acts with less information than the second.

Interpretation: Order matters because early investors act on less information. Later investors inherit the informational consequences of early actions or signals.

4.4 Theorem 3: simple path dependence converges to perfect communication

Theorem 3 states that when later investors can observe earlier investors' signals, the large-market outcome converges to the perfect-communication outcome.

Read:

- With many investors, the accumulated signal history becomes very informative.
- Later investors can infer the value precisely.
- In the limit, simple path dependence mimics full aggregation of signals.

Economic message: If signals are observable, sequential selling is less problematic in large markets. The major problem arises when investors observe actions but not signals.

4.5 Cascades

In the cascade regime, later investors observe only whether earlier investors bought or abstained. Once earlier actions are sufficiently informative, later investors may rationally ignore their own signals.

- A **positive cascade** occurs when subsequent investors buy regardless of their private signal.
- A **negative cascade** occurs when subsequent investors abstain regardless of their private signal.
- Cascades can begin after a small number of early decisions.

Interpretation: Cascades convert local early information into global offering outcomes. The market can stop aggregating information precisely when everyone is learning from everyone else.

4.6 Theorem 4: underpriced offerings can fail and overpriced offerings can succeed

Theorem 4 states that, with cascades, for any price $P > 1/3$, even infinitely many investors and infinitely many selling opportunities do not guarantee success of an underpriced issue. Conversely, for any price $P < 2/3$, an overpriced issue can still succeed perfectly.

Read:

- A few early unfavorable actions can trigger a negative cascade.
- Once a negative cascade begins, later investors abstain even if their private signals are favorable.
- A few early favorable actions can trigger a positive cascade.
- Once a positive cascade begins, later investors buy even if their private signals are unfavorable.

Economic message: Cascades make offering outcomes fragile. Correct pricing relative to true value is not enough: the sequence of early beliefs and actions can dominate fundamentals.

4.7 Theorem 5: risk-neutral issuers underprice to ensure full subscription

Theorem 5 states that an uninformed risk-neutral issuer optimally chooses the full-subscription price in the cascade setting.

Read:

- Full subscription is achieved by setting a low enough price that the first investor buys regardless of signal.
- This prevents a negative cascade from starting.
- Even a risk-neutral issuer accepts underpricing to avoid the risk of failure.

Economic message: The key force is not rationing or marginal-investor adverse selection. The issuer underprices because early refusal to buy can destroy demand for the whole issue.

4.8 Theorem 6: underpricing increases with uncertainty

Theorem 6 states that IPO underpricing increases when a mean-preserving spread is added to the common prior.

Interpretation: This connects the model to the empirical relationship between ex-ante uncertainty and IPO underpricing. Higher uncertainty makes early unfavorable actions more dangerous, so issuers underprice more aggressively.

4.9 Risk aversion and path-dependent pricing

The paper also asks whether changing the price in response to early sales could help the issuer. Theorem 7 says that for highly risk-averse issuers with many investors, it can be better to price low enough to create an immediate cascade than to vary price in response to early sales.

Economic message: Flexible pricing is not a complete solution. If the issuer is highly risk averse, the value of avoiding a negative cascade can dominate the value of adjusting price later.

4.10 Inside information and separating equilibria

The model is extended so the issuer observes a private signal. The issuer's signal is correlated with investors' signals because all signals are informative about the same underlying value.

In the example:

$$\Pr(LL | G) = \Pr(LH | G) = \Pr(HL | G) = \frac{1}{6}, \quad \Pr(HH | G) = \frac{1}{2}, \quad (6)$$

$$\Pr(HH | B) = \Pr(HL | B) = \Pr(LH | B) = \frac{1}{6}, \quad \Pr(LL | B) = \frac{1}{2}. \quad (7)$$

Here G is a good issuer with a high signal, and B is a bad issuer with a low signal.

Interpretation: A good issuer is more likely to face favorable investor signals. Therefore it can more safely choose a high price.

4.11 Proposition 1: separating equilibrium with simple path dependence

With simple path dependence, a separating equilibrium exists in which:

- the high-quality issuer prices at 0.75;
- the high-quality issuer fails completely with probability 1/3;
- the high-quality issuer succeeds partially with probability 1/6 and fully with probability 1/2;
- the low-quality issuer prices at 0.2, succeeds completely, and receives lower expected proceeds.

Economic message: Failure can be a cost of signaling. A high price is credible because a low-quality issuer is more likely to face unfavorable investor signals and therefore more likely to fail.

4.12 Proposition 2: cascades can improve the separating equilibrium

With cascades, a separating equilibrium exists in which:

- the high-quality issuer prices at 0.6 and expects to raise 0.9;
- the high-quality issuer fails completely with probability $1/6$;
- the high-quality issuer succeeds partially with probability $1/6$ and fully with probability $2/3$;
- the low-quality issuer prices at the full-subscription price of 0.25 and expects to raise 0.5.

Economic message: Cascades can help good issuers because they reduce the price needed for separation. This lowers failure risk and can make high-quality issuers better off.

5 Empirical implications and interpretation

5.1 Observed offering characteristics

The model predicts several observable patterns:

- Offerings can succeed or fail quickly because early purchases or abstentions can start cascades.
- Underpricing should be related to uncertainty.
- The role of the underwriter is partly to manage the sequence and communication structure of investors.
- A lack of early demand can be devastating, even if fundamentals are not bad.

Interpretation: The model provides a dynamic interpretation of roadshows, order books, early indications of interest, and withdrawn offerings.

5.2 Local versus national offerings

The paper argues that local offerings are more likely to allow investor communication and observation, while national or segmented offerings may reduce communication.

- Local offerings can be more vulnerable to negative cascades.
- National offerings can reduce the chance that early investor actions dominate the entire demand process.
- When inside information is important, national offerings are less likely to fail.

Economic message: Segmentation is valuable because it prevents early actions in one group from fully contaminating beliefs in all groups.

5.3 Inside information and offering failure

The inside-information extension implies that failed offerings need not be low-quality offerings.

- High-quality firms may set higher prices to separate.
- Higher prices create a risk of failure.
- Therefore some failed offerings can be high-quality issuers that drew unfavorable investor signals.

Interpretation: This interpretation is opposite to the naive view that failure always indicates low quality. In the model, failure can be the cost paid by good firms for signaling.

5.4 Underwriter incentives and pre-selling

The paper discusses pre-selling and underwriter incentives.

- Underwriters can line up investors before the offering becomes effective.
- If investors can observe who indicated interest, cascades can arise even in the pre-market.
- Underwriters may want to reduce investor communication, but they may also have incentives to advertise early success.
- Reputation and legal liability constrain false claims about demand.

Interpretation: The underwriter's job is not only to find investors, but also to control information flow during the book-building or selling process.

6 Tables and figures

6.1 Figures 1 and 2: simple path dependence

Read:

- Figure 1 shows the decision tree when two investors can observe prior signals.
- Investors decide whether to buy based on their own signal and earlier signals.
- Figure 2 shows issuer proceeds as a function of signal histories when the price is $P = 1/2$.
- In simple path dependence, outcomes depend on the ordering of H and L signals.

Economic message: These figures show that even without cascades, sequential selling creates path dependence because early investors act with less information than later investors.

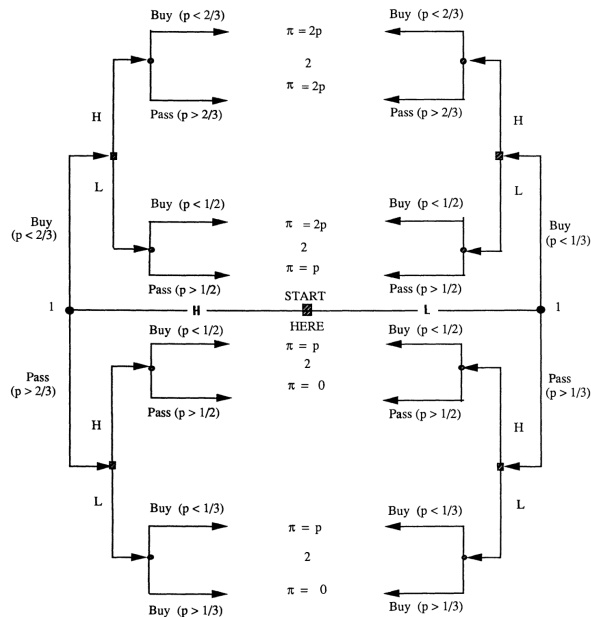


Figure 1. The simple path dependence game tree with two buyers. Square boxes indicate actions taken by nature, round boxes indicate decisions to be made by an investor. The tree shows that: (1) nature chooses the private signal (H or L) of investor 1; (2) investor 1 decides to *purchase* (which is optimal if the price is not above 0.33/0.67 if the signal is L/H) or to *pass* (which is optimal if the price is above 0.33/0.67 if the signal is L/H); (3) nature chooses the private signal of investor 2, (4) observing both his private signal and the signal of investor 1 (i.e., the node at which he is located in the game tree), investor 2 decides to purchase (which is optimal if the price is not above 0.25/0.5/0.75 and the two signals are LL/LH or HL/HH), or to

Figure 1: simple path-dependence game tree

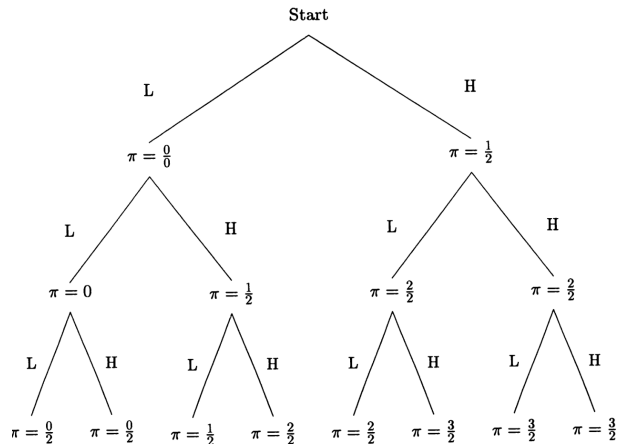


Figure 2. The simple path dependence equilibrium outcome tree when the price is $= 1/2$. This tree details the (cumulative) proceeds of an issuer (π) as a function of the signals (H or L) of the first three investors. For example, if these investors hold H , L , and L , respectively, no investors will purchase shares. Note that proceeds are path-dependent: an issuer receives one if the first three investors hold signals HLL than when they hold LLH .

Figure 2: simple path-dependence outcome tree

6.2 Figures 3 and 4: cascades

Read:

- Figure 3 modifies the game tree so investors observe prior actions rather than prior signals.
- Dotted lines show that the second investor cannot distinguish all signal histories once only actions are observed.
- Figure 4 shows that the tree can end in boxed nodes: once a cascade starts, later investors stop using their own information.
- Positive cascades lead all later investors to buy; negative cascades lead all later investors to abstain.

Economic message: These figures show why action-based learning is more fragile than signal-based learning. Observing actions can destroy information aggregation.

describes the decisions that the first two investors face in standard game tree format. Figure 3 differs from Figure 1 in that the two dotted lines indicate that the second investor does not know at which of the two nodes he moves: In the cascade scenario, he cannot observe the actual signal of his predecessor, only the purchasing decision. Like Figure 2, Figure 4 describes the equilibrium proceeds to the issuer as a function of the price and information

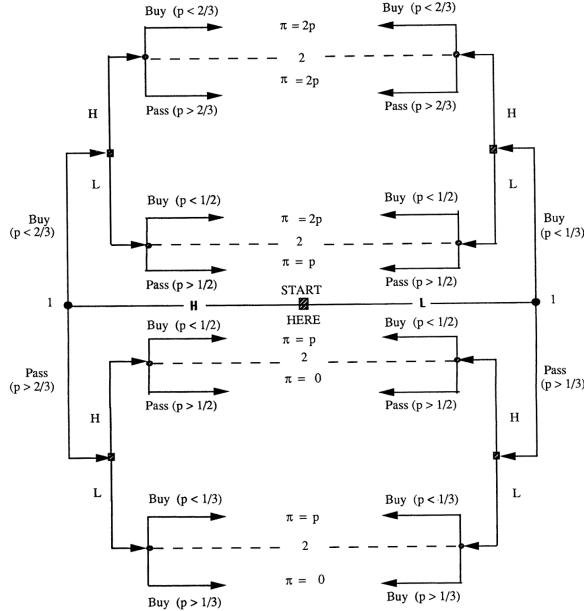


Figure 3. The cascades game tree with two buyers. For a general description, refer to the Figure 3: cascade game tree

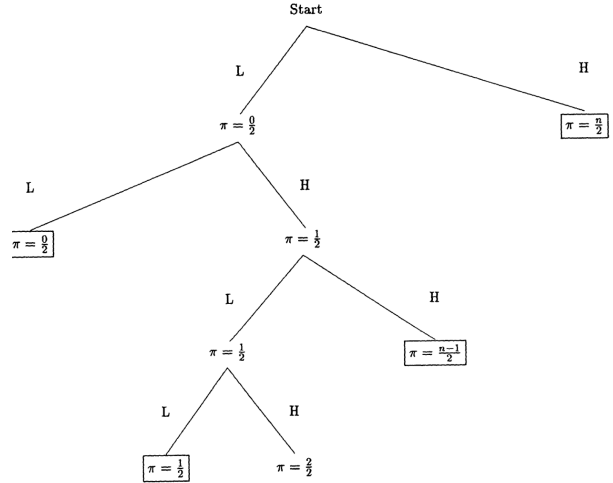


Figure 4. The cascade equilibrium outcome tree when the price is $P = 1/2$. This tree stails the (cumulative) proceeds of an issuer (π) as a function of the signals (H or L) of the first four investors. A boxed node indicates that all subsequent investors will act alike, and total proceeds can be determined (as a function of n the total number of investors). For example, if the

Figure 4: cascade outcome tree

6.3 Figure 5: expected proceeds across pricing regimes

Read:

- Figure 5 plots expected proceeds per investor as a function of the offer price.
- The perfect-communication and simple-path-dependence proceeds are smoother.
- Cascade proceeds are discontinuous and irregular because small price changes can change which histories trigger cascades.
- The figure shows why a risk-neutral issuer may choose a low full-subscription price.

Economic message: Cascades make demand extremely elastic and non-smooth. The issuer avoids failure by underpricing rather than by finely tuning price.

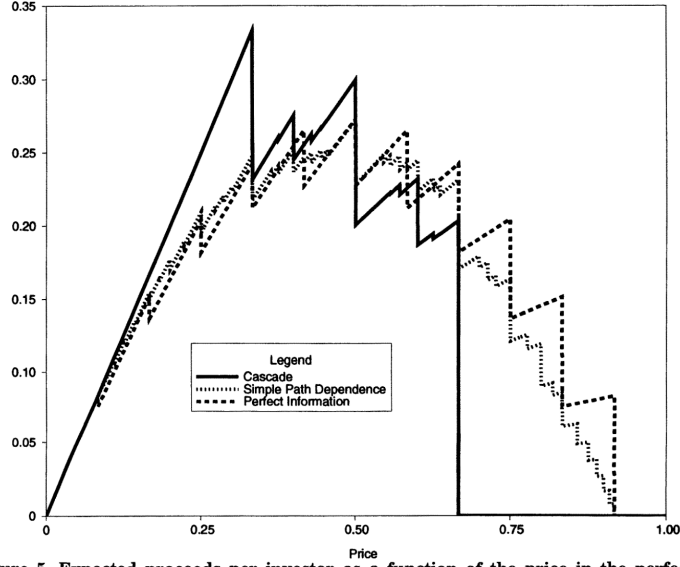


Figure 5. Expected proceeds per investor as a function of the price in the perfect communication, simple path dependence and cascades scenarios. The cascade proceeds are somewhat “chaotic,” with sharp irregular drops. At a price of $\frac{1}{3}$, the issuer maximizes expected proceeds (which are higher than those achieved in the other two scenarios). The perfect communication proceeds are quite regular and peak at a price of $\frac{1}{3}$. The simple path dependence proceeds track the perfect communication proceeds, albeit in a chaotic fashion similar to the cascades proceeds.

6.4 Table I: path-dependent pricing strategies under inside information

Read:

- Table I evaluates the inside-information example under simple path dependence.
- The low-quality issuer prefers the full-subscription price.
- The high-quality issuer can separate by setting a high price, but doing so creates substantial failure risk.
- The table supports Proposition 1: G prices at 0.75 and B prices at 0.20.

Economic message: A high price can credibly signal high quality because low-quality issuers face a higher probability of unfavorable outside information.

Table I
The Path-Dependent Pricing Strategies under Inside Information

This table details investors' behavior and the issuer's expected proceeds (from both investors) at all potentially profit-maximizing prices, given the true and given the inferred type of the firm. In each panel, the second line lists all possible distributions of outside signals. The third line lists the true probability of each firm to be faced with a particular distribution of outside signals. Subsequent lines list—for all prices—the optimal decisions of the two investors (given the inferred type and observed signal(s)); the expected share value is always $(k+1)/(n+2)$ where k is the number of H signals in n draws. To illustrate, “buy/abstain” means that the first investor optimally decides to purchase and the second investor optimally decides to abstain. The expected proceeds of the issuer, in column 6, are obtained by multiplying each state probability by the number of investors purchasing and the assumed issue price, and summing over all states. From the table, the separating equilibrium is for B to price at $\frac{1}{5}$ and G to price at $\frac{3}{4}$.

True and Inferred Firm Value	B (Issuer holds one L signal)				E Proc
	L/L	L/H	H/L	H/H	
	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	
Optimal Investor Behavior					
Price = $\frac{1}{6}$	buy/buy	buy/buy	buy/buy	buy/buy	0.400
Price = $\frac{1}{4}$	buy/abstain	buy/buy	buy/buy	buy/buy	0.375
Price = $\frac{1}{3}$	abstain/abstain	abstain/buy	buy/buy	buy/buy	0.277
Price = $\frac{2}{5}$	abstain/abstain	abstain/buy	buy/buy	buy/buy	0.333
Price = $\frac{1}{2}$	abstain/abstain	abstain/abstain	buy/buy	buy/buy	0.250
Price = $\frac{3}{5}$	abstain/abstain	abstain/abstain	abstain/abstain	abstain/buy	0.100
Price = $\frac{2}{3}$					
Price = $\frac{3}{4}$					
Price = $\frac{4}{5}$					

True and Inferred Firm Value Investors Observe: True Probability:	G (Issuer holds one H signal)				E Proc
	L/L	L/H	H/L	H/H	
	1/6	1/6	1/6	1/2	
Optimal Investor Behavior					E Proc
Price = 1/6					
Price = 1/4					
Price = 1/3					
Price = 2/5	buy/buy	buy/buy	buy/buy	buy/buy	0.800
Price = 1/2	buy/abstain	buy/buy	buy/buy	buy/buy	0.917
Price = 3/5	abstain/abstain	abstain/buy	buy/buy	buy/buy	0.900
Price = 2/3	abstain/abstain	abstain/abstain	buy/abstain	buy/buy	0.778
Price = 3/4	abstain/abstain	abstain/abstain	buy/abstain	buy/buy	0.875
Price = 4/5	abstain/abstain	abstain/abstain	abstain/abstain	abstain/buy	0.400
False but Inferred Firm Value					
Inferred Firm Value Investors Observe: True Probability:	G (Issuer falsely believed to hold one H signal)				E Proc
	L/L	L/H	H/L	H/H	
	1/2	1/6	1/6	1/6	
Optimal Investor Behavior					E Proc
Price = 1/6					
Price = 1/4					
Price = 1/3					
Price = 2/5	buy/buy	buy/buy	buy/buy	buy/buy	0.800
Price = 1/2	buy/abstain	buy/buy	buy/buy	buy/buy	0.750
Price = 3/5	abstain/abstain	abstain/buy	buy/buy	buy/buy	0.500
Price = 2/3	abstain/abstain	abstain/abstain	buy/abstain	buy/buy	0.333
Price = 3/4	abstain/abstain	abstain/abstain	buy/abstain	buy/buy	0.375
Price = 4/5	abstain/abstain	abstain/abstain	abstain/abstain	abstain/buy	0.133

6.5 Table II: cascade pricing strategies under inside information

Read:

- Table II repeats the inside-information example under cascades.
- The low-quality issuer's full-subscription strategy becomes more attractive.
- The high-quality issuer can separate at a lower price than in Table I.
- The table supports Proposition 2: G prices at 0.6 and expects higher proceeds with lower failure risk.

Economic message: Cascades are not always bad for issuers. In the inside-information example, cascades relax the signaling problem and reduce the price needed for separation.

Table II
The Cascade Pricing Strategies under Inside Information

This table details investors' behavior and the issuer's expected proceeds (from both investors) at all potentially profit-maximizing prices, given the true and given the inferred type of the firm. In each panel, the second line lists all possible distributions of outside signals. The third line lists the true probability of each firm to be faced with a particular distribution of outside signals. Subsequent lines list—for all prices—the optimal decisions of the two investors (given the inferred type and observed signals); the expected share value is always $(k+1)/(n+2)$ where k is the number of H signals in n draws. To illustrate, "buy/abstain" means that the first investor optimally decides to purchase and the second investor optimally decides to abstain. The expected proceeds of the issuer, in column 6, are obtained by multiplying each state probability by the number of investors purchasing and the assumed issue price, and summing over all states. From the table, the separating equilibrium is for B to price at $1/4$ and G to price at $2/5$.

True and Inferred Firm Value	B (Issuer holds one L signal)				E Proc
	L/L	L/H	H/L	H/H	
	$1/2$	$1/6$	$1/6$	$1/6$	
	Optimal Investor Behavior				
Price = $1/5$	buy/buy	buy/buy	buy/buy	buy/buy	0.400
Price = $1/4$	buy/buy	buy/buy	buy/buy	buy/buy	0.500
Price = $1/3$	abstain/abstain	abstain/buy	buy/buy	buy/buy	0.277
Price = $2/5$	abstain/abstain	abstain/abstain	buy/abstain	buy/buy	0.333
Price = $1/2$	abstain/abstain	abstain/abstain	abstain/abstain	abstain/abstain	0.250
Price = $3/5$	abstain/abstain	abstain/abstain	abstain/abstain	abstain/abstain	0.000

Table II – Continued

True and Inferred Firm Value	G (Issuer falsely believed to hold one H signal)				E Proc
	L/L	L/H	H/L	H/H	
	$1/2$	$1/6$	$1/6$	$1/6$	
	Optimal Investor Behavior				
Price = $1/5$	buy/buy	buy/buy	buy/buy	buy/buy	0.800
Price = $1/4$	buy/buy	buy/buy	buy/buy	buy/buy	1.000
Price = $1/3$	abstain/abstain	abstain/abstain	buy/abstain	buy/buy	0.900
Price = $2/5$	abstain/abstain	abstain/abstain	buy/abstain	buy/buy	0.778
Price = $1/2$	abstain/abstain	abstain/abstain	abstain/abstain	abstain/abstain	0.875
Price = $3/5$	abstain/abstain	abstain/abstain	abstain/abstain	abstain/abstain	0.000

7 Short summary

- Welch studies IPOs sold sequentially to investors who learn from earlier purchase decisions.
- Later investors can use early investor behavior as information about the offering.
- If investors observe prior signals, simple path dependence arises and converges to perfect communication in large markets.
- If investors observe only prior actions, cascades can occur.
- Cascades cause later investors to ignore their private signals.
- Under cascades, underpriced issues can fail and overpriced issues can succeed.
- A risk-neutral uninformed issuer may optimally underprice to guarantee early buying and avoid a negative cascade.
- This mechanism makes the winner's curse largely irrelevant for the pricing result.
- With issuer inside information, high-quality issuers may signal by setting high prices and risking failure.
- Underwriters may have a useful role in segmenting markets and managing investor communication.
- The incremental contribution is to connect informational cascades to IPO underpricing, issue failure, and underwriter distribution strategy.

8 Conclusion

8.1 Core conclusion

Welch shows that sequential sales can generate informational cascades in IPO markets. Later investors learn from the purchase decisions of earlier investors, and once a cascade begins, later investors ignore their private information. This dynamic can make IPO demand extremely path dependent.

8.2 Economic intuition

The intuition is that early investors create a public signal through their actions. If early investors buy, later investors infer that early signals were favorable. If early investors abstain, later investors infer that early signals were unfavorable. Because each later investor also contributes to the public history, the process can rapidly become self-reinforcing.

8.3 Incremental contribution takeaway

The paper's incremental contribution is to show that IPO underpricing and offering failure can be explained by sequential learning and cascades rather than only by rationing or the winner's curse. It also explains why the organization of the selling process matters: underwriters can affect outcomes by controlling the order of investor contact and the flow of information among investors.

8.4 Broader takeaway

The paper turns IPO demand into a dynamic social-learning problem. The success of an offering depends not only on fundamentals and price, but also on who is approached first, what later investors can observe, and how easily investor beliefs can cascade.